

Siyuan Liang

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EDUCATION

University of Science and Technology of China (USTC)

Sep 2023–Jun 2027 (expected)

School of the Gifted Young

B.Sc. in Mathematics and Applied Mathematics, Hua Luogeng Class in Mathematics

Minor in Computer Science and Technology

- **Academic standing:** Top 25% in the School of the Gifted Young.
- **Honors:** First Prize, Chinese Mathematics Competitions (CMC), Anhui Division (Mathematics Majors), 2024; First Prize, Chinese National High School Mathematics League, 2022.
- **Selected course grades:** Linear Algebra A2 (96), Advanced Mathematical Statistics (95), Functional Analysis (93), Mathematical Analysis A1 (91).

Doctoral admission, USTC

Admitted; enrollment pending

Planned joint doctoral training at Shenzhen Loop Area Institute (SLAI).

Prospective supervisor: **Difan Zou**, Assistant Professor, School of Computing and Data Science, The University of Hong Kong.

RESEARCH INTERESTS

Theoretical machine learning; deep learning; theoretical questions in modern artificial intelligence; AI for Mathematics.

SELECTED PROJECTS

Operations Research and Optimization Algorithms

Individual project

- Implemented the simplex method, Dijkstra's algorithm, quasi-Newton methods, and FISTA, including line-search and restart strategies for FISTA.
- Analyzed FISTA's convergence properties and compared strategies through numerical experiments.

Reinforcement Learning for Gomoku

Team project

- Implemented a neural-network policy in Python and PyTorch, designed the reward function, and trained a reinforcement-learning model for Gomoku.
- Received a score of 195/200 in the final project assessment.

Football Match Prediction with Linear Algebra

Individual project

- Developed a method to predict missing football match outcomes and analyzed its properties using linear algebra and graph theory.

TEACHING EXPERIENCE

Teaching Assistant, Mathematical Analysis B1

Sep 2026–present (scheduled through Jan 2027)

- Prepare exercise solutions and mathematical proofs in LaTeX; support undergraduates in calculus and series, with supplementary topics in real analysis, complex analysis, and differential geometry.

Teaching Assistant, Linear Algebra B1

Sep 2025–Jan 2026

- Prepared exercise solutions and mathematical proofs in LaTeX; supported undergraduates in matrix theory and vector spaces.

TECHNICAL SKILLS & LANGUAGES

Programming and tools: Python, PyTorch, NumPy, pandas, Git, Jupyter Notebook; C, R, LaTeX.

English: College English Test, Band 6 (CET-6): 556.